



Pharm*Achieve*

HEALTH PROMOTION AND IMMUNIZATIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

Anti-HBs Antibody to Hepatitis B Surface Antigen

BCGP Board Certified Geriatric Pharmacist

CDE Certified Diabetes Educator

DTaP Diphtheria Tetanus Acellular Pertussis

GLP-1 RA Glucagon-Like Peptide-1 Receptor Agonist

HBV Hepatitis B Virus

HCV Hepatitis C Virus

Hib Haemophilus Influenzae Type B

HPV Human Papillomavirus

ICU Intensive Care Unit

ID Intradermal

IIV3 Trivalent Inactivated Influenza Vaccine

IIV4 Quadrivalent Inactivated Influenza Vaccine

IM Intramuscular

IPV Inactivated Poliomyelitis Vaccine

IV Intravenous

LAIV Live Attenuated Influenza Vaccine

LRT Lower Respiratory Tract

MMR Measles, Mumps, Rubella

MMRV Measles, Mumps, Rubella, Varicella

NACI National Advisory Committee on Immunization

PHAC Public Health Agency Of Canada

PNEU-C-13 Pneumococcal Conjugate 13-Valent

PNEU-P-23 Pneumococcal Polysaccharide 23-Valent

PO By Mouth, Orally

RSV Respiratory Syncytial Virus

SC Subcutaneous

STI Sexually Transmitted Infection

TB Tuberculosis

Td Tetanus Diphtheria

Tdap Tetanus Diphtheria Acellular Pertussis

WHAT IS HEALTH PROMOTION?

Health Promotion:

- Social and environmental interventions → enable people/communities to **improve their health** and have **increased control over their health**
- Examples:
 - Promoting behaviours that are healthy, establishing healthy public policies, creating environments that are supportive, and strengthening personal skills
 - Preventative care: vaccines, Point-of-Care testing...etc.

WHAT IS HEALTH PROMOTION?

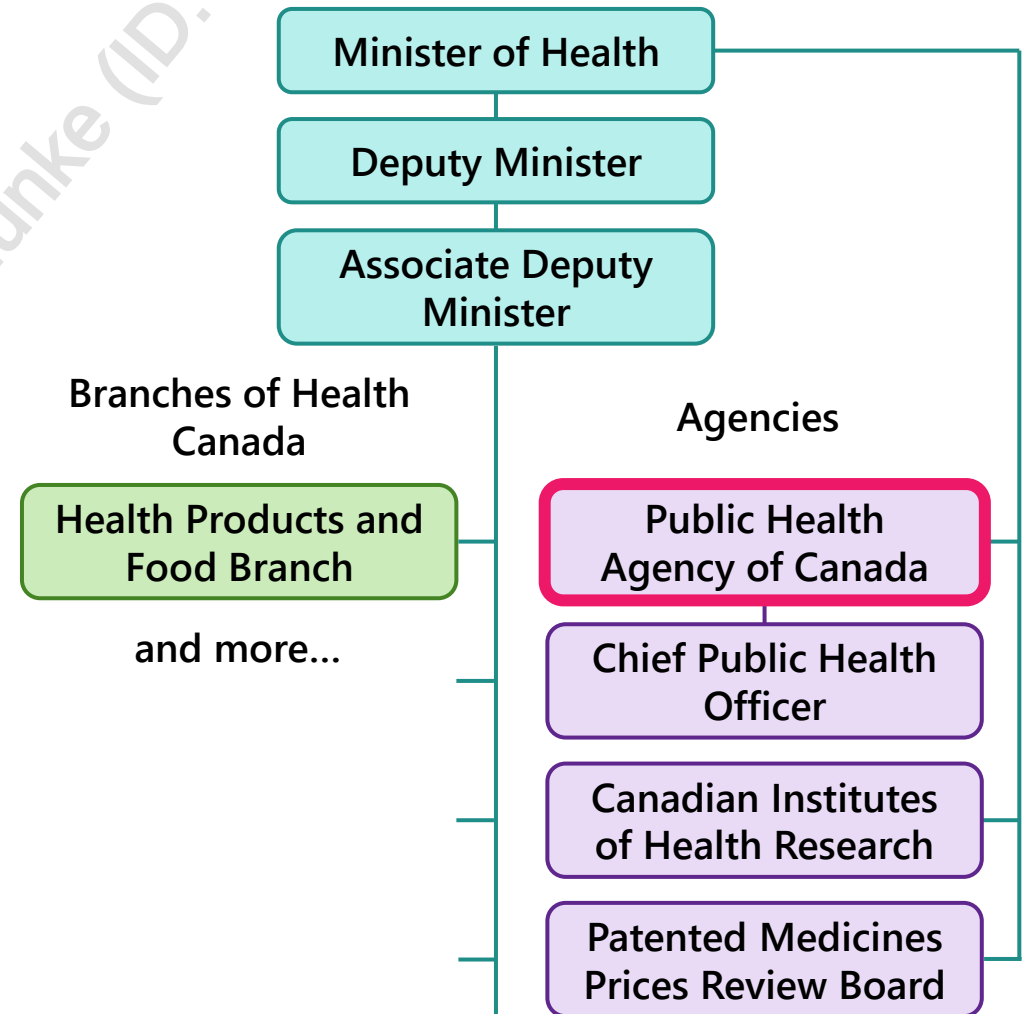


The process of enabling people to increase control over, and to improve, their health

– *World Health Organization*

WHAT IS THE ROLE OF THE PUBLIC HEALTH AGENCY OF CANADA?

- To protect the health and safety of all Canadians by:
 - Focusing on preventing diseases and injuries
 - Responding to public health threats (e.g. emergencies and infectious outbreaks)
 - Promoting physical and mental health
- Involves collaboration of Federal, Provincial/Territorial and municipal governments



WHAT ARE THE DETERMINANTS OF HEALTH?

Conditions in which we are born and raised, live, and work which impact our health and well-being



Childhood experiences



Housing



Education



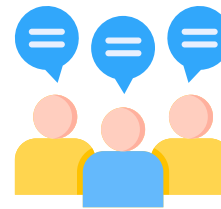
Social support



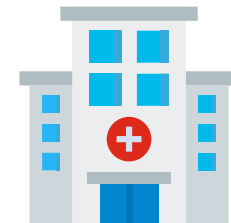
Family income



Employment



Our communities



Access to health services

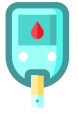
HOW CAN PHARMACISTS IMPLEMENT HEALTH PROMOTION SERVICES INTO THEIR PRACTICE?



Community outreach programs



Interprofessional collaboration



Diabetes care (CDE)



Geriatric care (BCGP)



Travel consultation



Vaccinations



Smoking cessation



Medication reviews



Pharmaceutical opinions



Point-of-care testing



Cardiovascular clinic



Skin/sun care clinic



Pharmacogenetic Screening



Cannabis consults



Naloxone kits



Pharmacist prescription renewals and adaptations

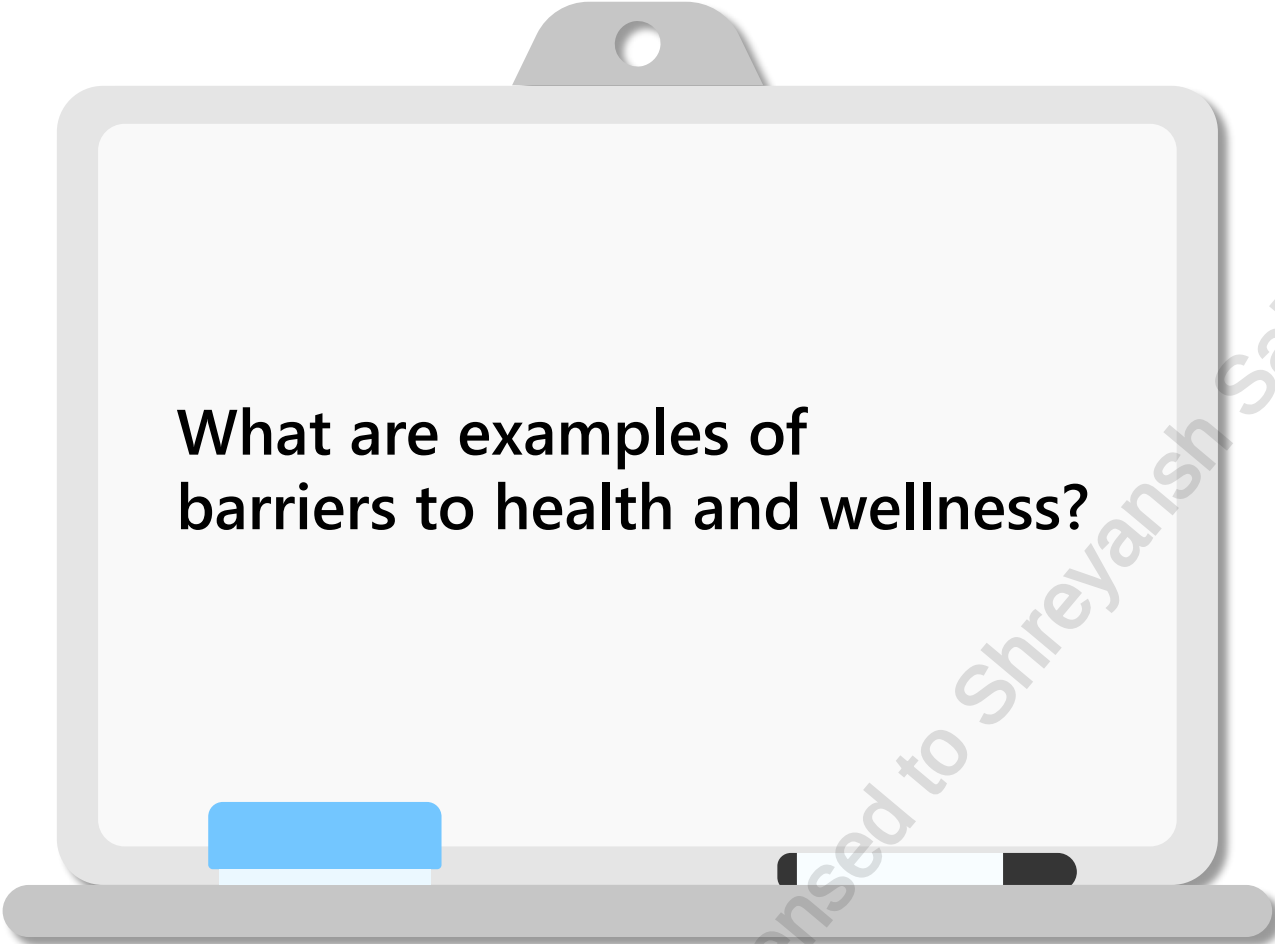


Minor ailments

CANADA AND HEALTH PROMOTION



DISCUSSION



**What are examples of
barriers to health and wellness?**

IMMUNIZATIONS - ROUTES OF ADMINISTRATION

There are 5 routes of administration for injectable drugs:

Intramuscular (IM): injection into muscle tissue

- Most vaccines and vitamin B12
- Of the non-IV parenteral routes, this has the fastest onset because muscle tissue is highly vascular

Subcutaneous (SC): injection into tissue below the dermis of the skin

- Insulin, GLP-1 RA, most biologic drugs and some vaccines are administered in this way
- Generally, has a slower onset of action than IM injections

Intradermal (ID): injection directly into the dermis of the skin

- This is often used for skin reaction tests (e.g. tuberculosis skin test)

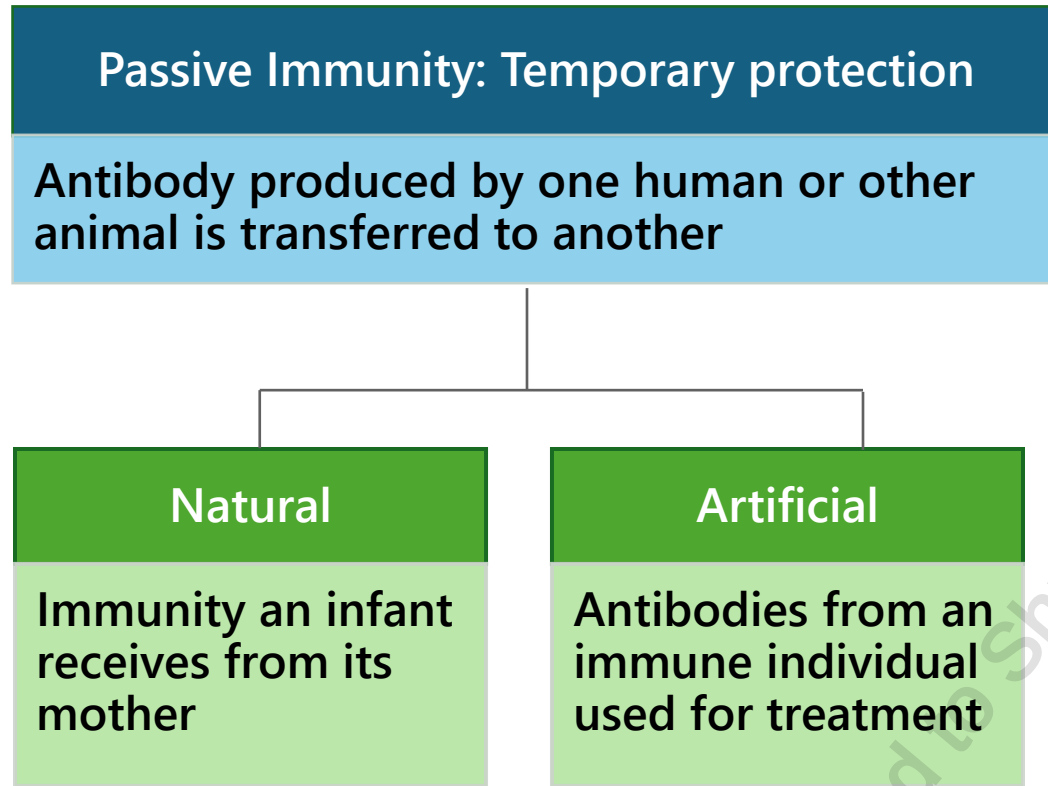
Intravenous (IV): injection directly into the venous blood

- Many hospital drugs are administered in this way
- This route of injection will not be covered in this course

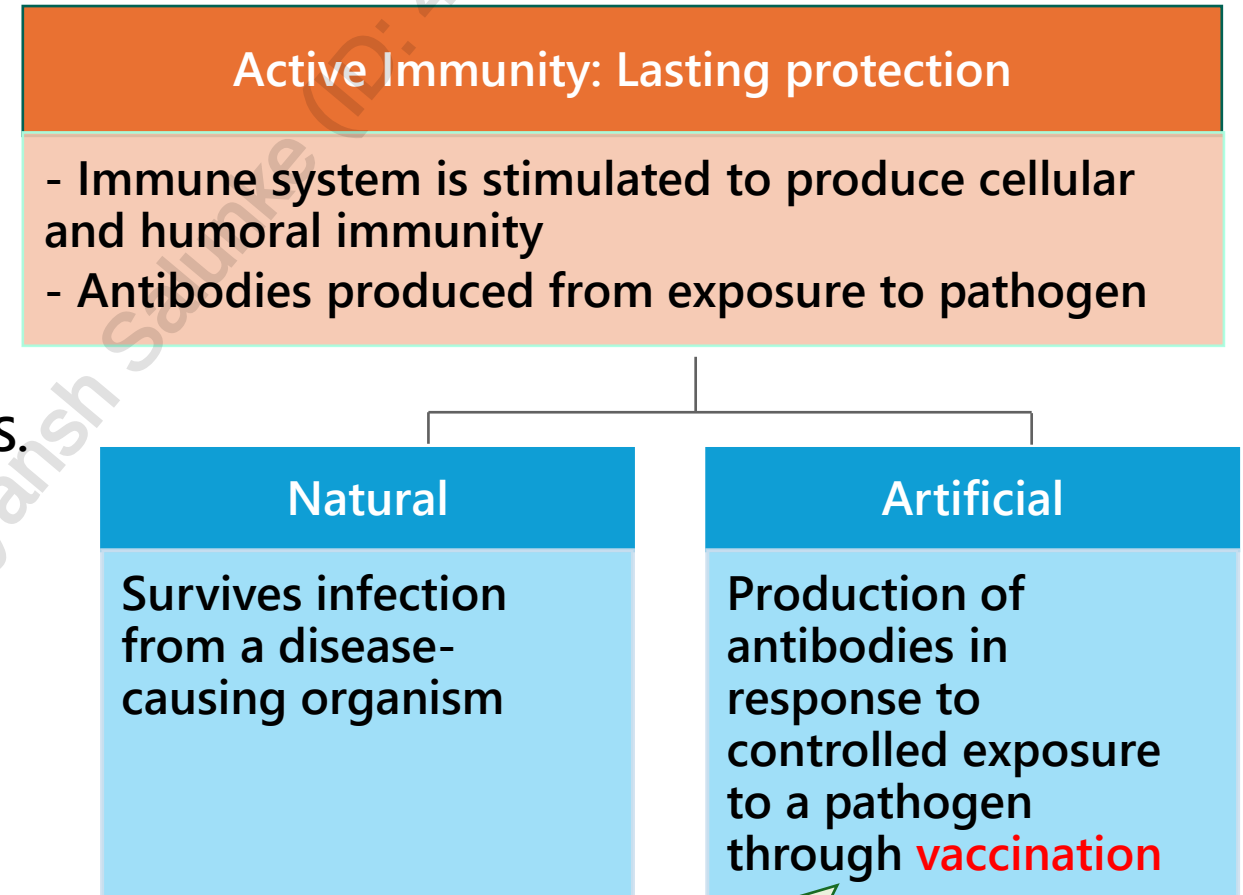
Intranasal: drug administered in one or both nostrils

- Examples: FluMist® (nasal spray influenza vaccine), naloxone nasal spray
- This route is not very common

TYPES OF IMMUNITY: REVIEW OF VACCINES



VS.



What is the difference between live and inactivated vaccines?

MECHANISM OF ACTION

- Vaccinations produce active immunity
- Immune response produced is similar to that caused by a natural infection
- Vaccinations produce immunologic memory that can last for many years
- Protection comes without complications/risks of actual disease
- There are two types of vaccine formulations:

Live attenuated	Inactivated
• Live bacterium or virus in a (weakened) form • Agent must replicate to be effective • In immunodeficient patients uncontrolled replication can cause severe disease - AVOID • Has a <u>cellular</u> and <u>humoral</u> immune response	• Inactivated form or antigenic fragments • Agent is not live so it cannot replicate • Does not cause disease, even in immunodeficient patients - SAFE • Immune response is <u>primarily humoral</u> • Always requires more than one dose

EXAMPLES OF VACCINES

- Severe asthma: defined as currently on oral or high dose inhaled glucocorticosteroids or active wheezing

Vaccine Type	Vaccine Examples	Contraindications
LIVE, ATTENUATED VACCINES	• Bacillus Calmette-Guérin (BCG) • Measles • Mumps • <i>Rotavirus</i> • Live attenuated influenza (intranasal influenza vaccine) • Rubella • Smallpox • Typhoid (oral) • Varicella (chickenpox) • Yellow fever	• Anaphylactic reaction to vaccine components • <i>Severe asthma OR* medically attended wheezing 7 days prior to vaccination</i> • <i>Rotavirus: pts with congenital GI malformation or history of intussusception</i> • Immunocompromised patients • Pregnancy • MMR, MMRV, BCG in pts with active untreated TB
INACTIVATED VACCINES	• Acellular pertussis • Cholera & travelers' diarrhea • Diphtheria toxoid • H. influenzae type b • Hepatitis A, Hepatitis B • Human papillomavirus • Inactivated herpes zoster • Inactivated poliomyelitis • Japanese encephalitis • Meningococcal • Pneumococcal • Rabies • Tetanus toxoid • Inactivated influenza • Typhoid (injectable)	• Anaphylactic reaction to vaccine components

VACCINATION SCHEDULES

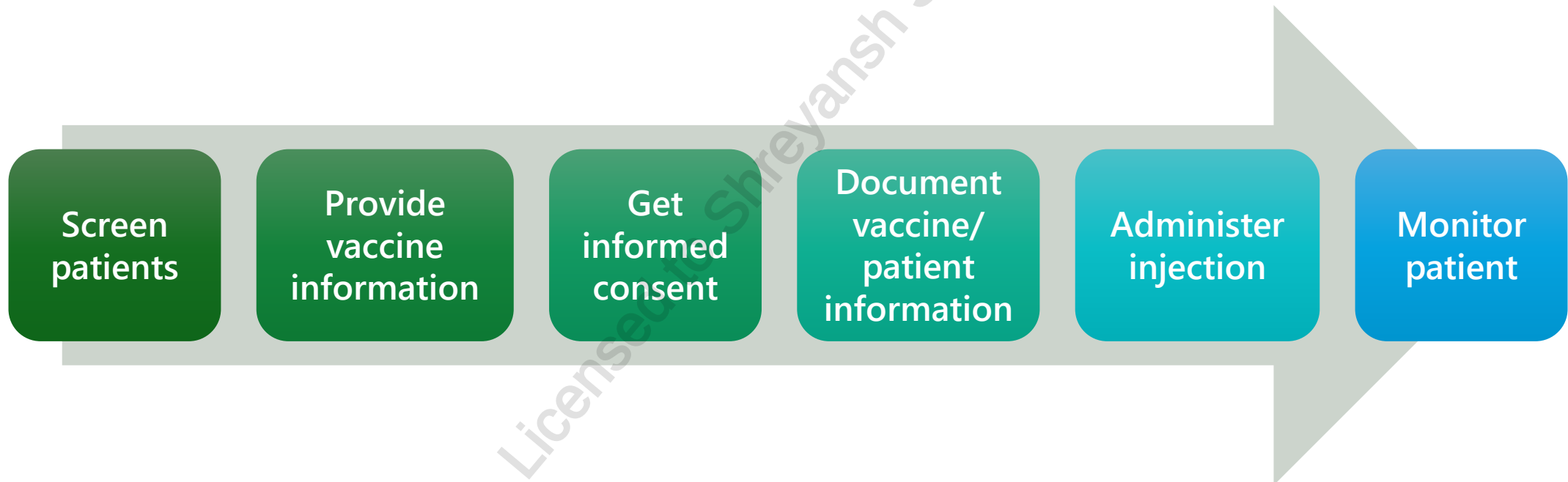
Use the following two resources to review provincial and federal recommendations:

- Provincial and territorial health agencies have created vaccination schedules which can be accessed using the following online tool: <http://healthycanadians.gc.ca/apps/schedule-calendrier/index-eng.php>
- Health Canada has published a summary of the current routine vaccination schedule for infants and children in all provinces and territories: <https://www.canada.ca/en/public-health/services/provincial-territorial-immunization-information/provincial-territorial-routine-vaccination-programs-infants-children.html> (Last update October 2024)
- Canadian Immunization Guide has a section on general recommendations for immunization schedules: <https://www.canada.ca/en/public-health/services/canadian-immunization-guide.html>

IMMUNIZATION PROCEDURE

All immunizations should follow the 6 steps outlined below to:

- Minimize the risk of severe reactions or unintended consequences
- Ensure patient understands the risks, benefits, and side effects of the vaccine
- Fulfill documentation requirements



IMMUNIZATION PROCEDURE

- **Provide information**
 - Benefits and risks of the vaccine (risk of disease in absence of vaccination)
 - Common side effects, contraindications, myths, and misconceptions
- **Obtain informed consent**
 - Immunization CANNOT be given without patient's informed consent
 - Informed consent is **NOT** required for life-saving injections (e.g. EpiPen®, naloxone)
 - If a child is to be vaccinated, parent consent should be obtained unless immunizer believes child is capable of understanding benefits and risks of vaccination
- **Document**
 - Product name, manufacturer and lot number, route of administration, indication, patient information, immunizer, date/time administered, post-injection adverse effects and management, follow-up

PREPARING TO ADMINISTER

- Choose correct needle LENGTH and GAUGE to minimize tissue injury and patient discomfort
 - Note: the **higher** the gauge, the **smaller** the opening of the needle
- Use a longer needle if there is more SC fat

Route	Needle gauge	Needle length (inches)
Subcutaneous	25	Adult: 5/8 inch Child: 5/8 inch
Intramuscular	22-25	Deltoid muscle Adults and adolescents (>12 years): 5/8 – 1 ½ inch* Child: 5/8 – 1 inch

*A 5/8" to 1" needle may be used if <130 lbs. In males weighing 130-260 lbs and females weighing 130-200 lbs, a 1" needle is recommended. In males weighing >260 lbs and females weighing >200 lbs, a 1 ½" needle is recommended

ADMINISTRATION & FINAL STEPS

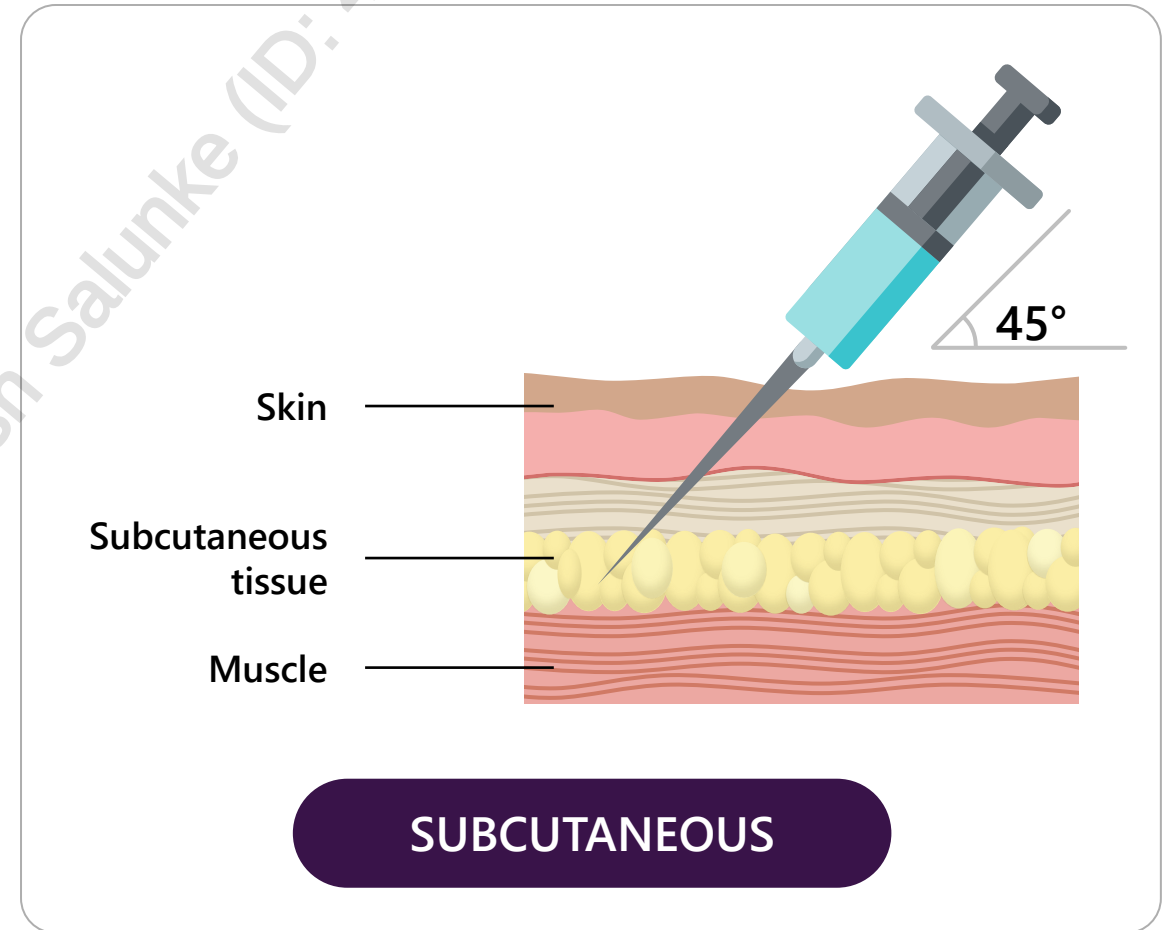
- Select site free from infection and injury
- Cleanse area with alcohol swab and allow the alcohol to dry
- Target area by gently stretching the skin (IM) or pinching the skin (SC)
- Use a dart-like motion with steady pressure
- Inject at:
 - 90° angle for IM
 - 45° angle for SC
- Have patient remain under surveillance for approximately 15 minutes after the injection

Multiple injections into the same area should be separated by at least 1 inch

SUBCUTANEOUS INJECTIONS

INJECTION SITE

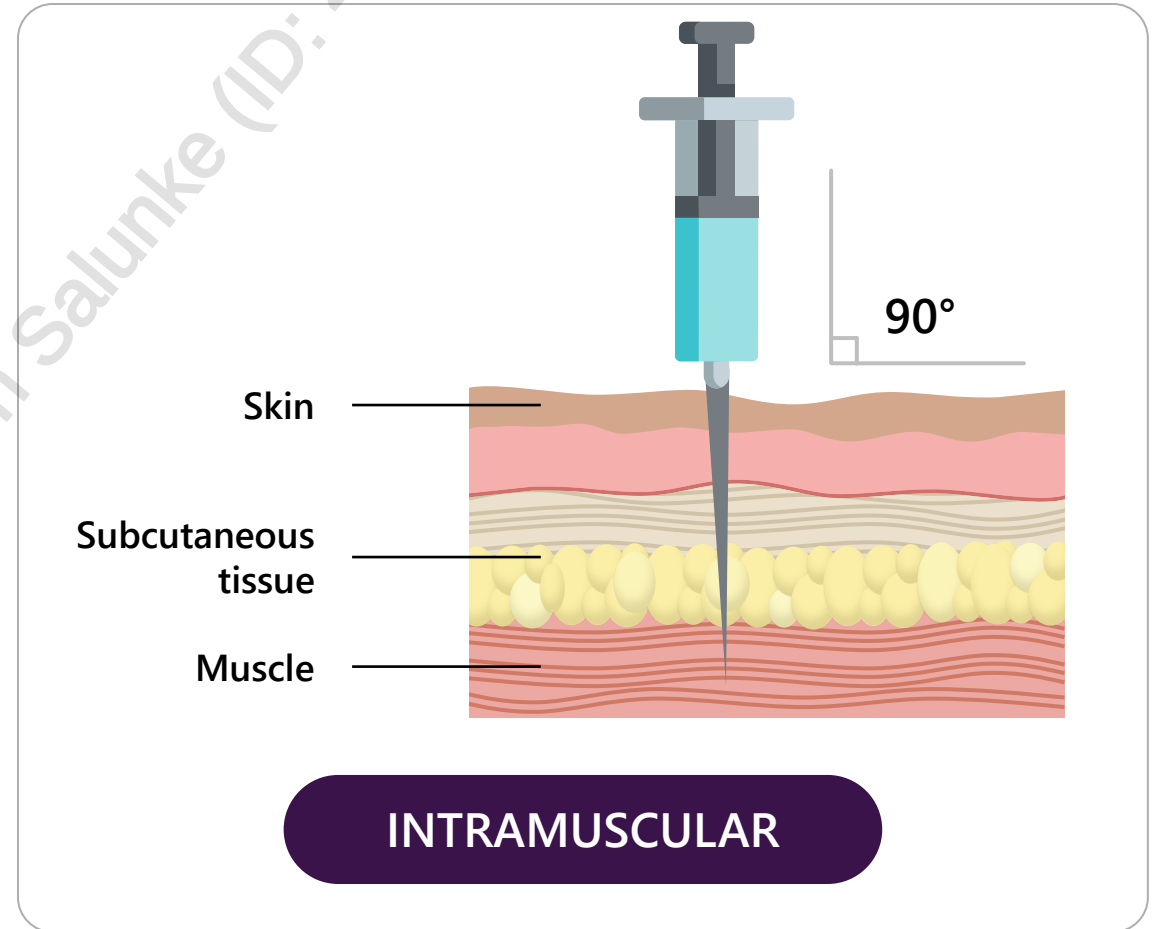
- Generally:
 - For children under 12 months of age, use the fatty tissue overlying the ***anterolateral thigh***
 - For children *12 months or older* and adults, use the fatty tissue overlying ***the upper triceps area or the anterolateral thigh***



INTRAMUSCULAR INJECTIONS

INJECTION SITE

- Generally:
 - For children under 12 months of age: *anterolateral thigh muscle*
 - For children aged 12 months to 12 years: *deltoid muscle of the arm* or *the anterolateral thigh muscle*
 - For adolescents and adults: *deltoid muscle of the arm preferred*

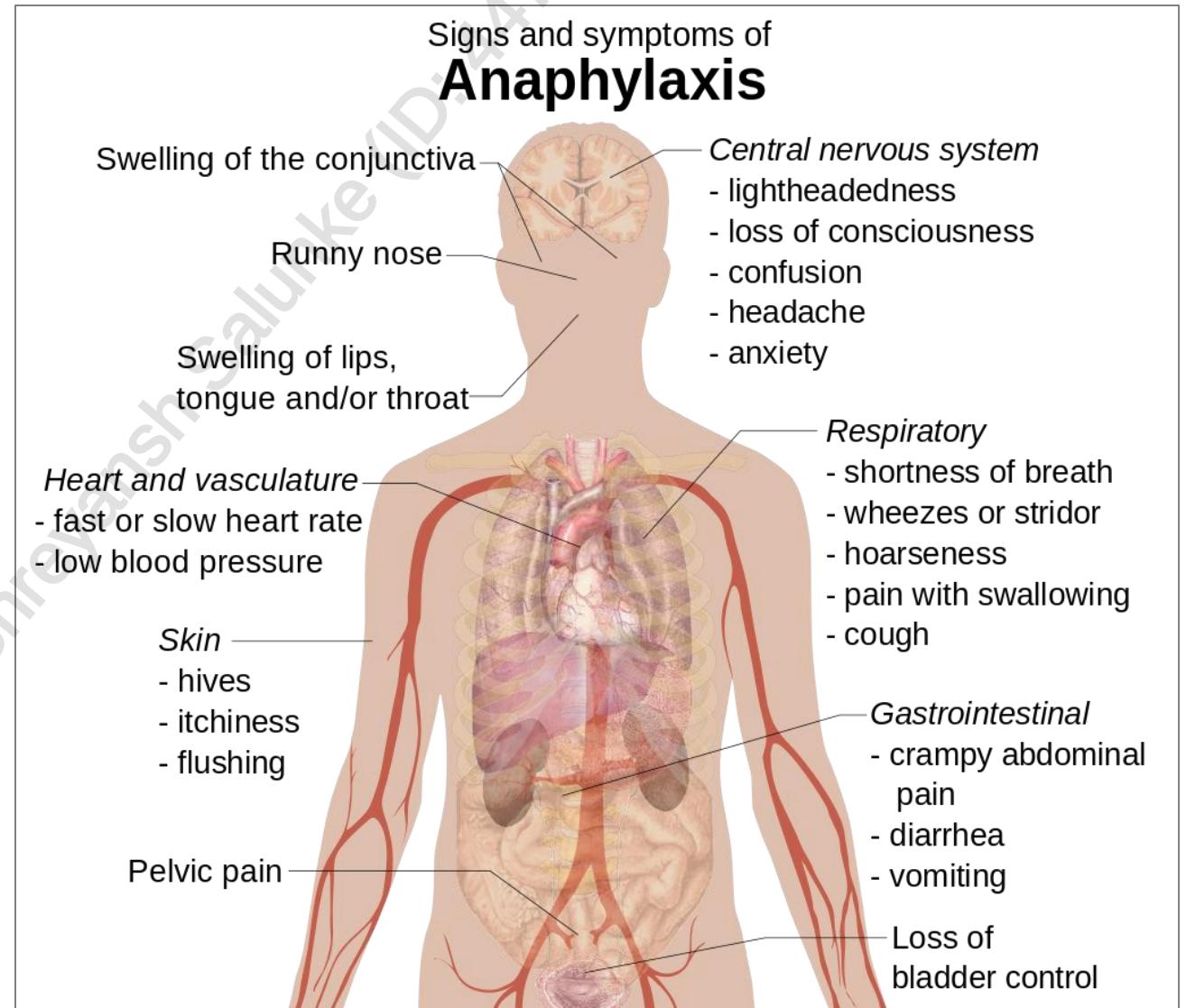


MONITORING FOR REACTIONS

1. Before releasing patient, **monitor for at least 15 minutes** for immediate adverse reactions
2. Two of the more serious immediate reactions are **syncope (anxiety-related) and anaphylaxis**
3. Other adverse reactions such as injection site pain, redness and swelling are more common and are delayed onset reactions
4. Adverse reactions that you become aware of should be reported via Canadian Adverse Events Following Immunization Surveillance System (CAEFISS) (a public health post-market vaccine safety monitoring system run by Health Canada which can be accessed via a web site)

ANAPHYLAXIS

- Anaphylaxis is a life-threatening allergic reaction involving multiple organ systems
- Symptoms can progress rapidly and may be life-threatening if airway obstruction or a severe drop in blood pressure occurs
- Most effective treatment for anaphylaxis is epinephrine administered as an IM injection in the thigh



NEEDLE-STICK INJURY PREVENTION

- ✓ Infection control guidelines are designed to protect providers and patients from exposure to diseases spread by blood and bodily fluids
- ✓ Providers should treat blood and other bodily fluids of ALL patients as a potential source of infectious disease
- ✓ The following precautions should be taken:
 - ✓ Wear gloves if likely to touch body substances or mucous membranes
 - ✓ Wear eye protection and face mask if likely to be splashed
 - ✓ Wear lab coat or gown if likely to be soiled
 - ✓ Place needles in sharps containers (do NOT recap)
 - ✓ Remove gloves using aseptic technique and wash hand immediately
 - ✓ Place soiled articles in plastic bag for disposal
 - ✓ Place soiled linen in laundry bag



STORING VACCINES/BIOLOGICS

- To protect against loss of efficacy, vaccines, and biologics must be transported and stored in a cold environment with specific temperature requirements called **COLD CHAIN**
- The cold chain is strictly monitored and regulated by monitoring refrigerator temperatures and reporting cold chain incidents to public health unit
- Pharmacies must have medication storage refrigerators that meet standards of temperature consistency and maintaining a record of the temperature is required
- GENERAL GUIDELINES
 - Liquid vaccines and reconstituted products must be kept at **+2°C to +8 °C**
 - Reconstituted products must be used in the timeframe specified by the manufacturer
 - Frozen products must be stored between **-15 °C to -50 °C**

EXAMPLES OF VACCINATIONS BY PHARMACISTS



NEW: COVID-19 Vaccine

• Hepatitis A	• Pneumococcal disease
• Hepatitis B	• Tuberculosis
• Herpes Zoster	• Rabies
• Haemophilus Influenza Type B	• Varicella Virus
• Meningitis	• Typhoid Disease
• Human Papillomavirus	• Yellow Fever
• Japanese Encephalitis	• Influenza
• Respiratory Syncytial Virus (RSV)	

HEPATITIS A

DISEASE SYMPTOMS AND COMPLICATIONS	Viral infection of the liver; flu-like symptoms (fatigue, fever), GI symptoms, pale stools, jaundice, paralysis, death (rare)
DISEASE SPREAD	• Contaminated food or water (i.e. fecal-oral route) • Sexual contact, needle sharing • Hand-to-mouth contact, person-to-person contact
VACCINE INDICATION	Pre-exposure immunization in those 6 months of age and older (e.g. protection against virus when travelling to endemic countries)
VACCINE ADVERSE EFFECTS	Local reaction (soreness, swelling, redness at injection site)
DOSING SCHEDULE	2 doses: initial (ideally 2 weeks before travel) then booster dose at 6-36 months afterwards (depending on product)

HEPATITIS B

DISEASE SYMPTOMS AND COMPLICATIONS	Viral infection of the liver; same symptoms as hepatitis A for acute infection - chronic infection can cause liver cancer, cirrhosis, death
DISEASE SPREAD	• Mother to newborn during delivery • Contact with blood or bodily fluids (e.g. sexually transmitted)
VACCINE INDICATION	• Routine immunization in children • Pre-exposure immunization for high-risk groups (e.g. protection against virus when travelling to endemic areas, injection drug users, men who have intercourse with men, health care workers at risk of bodily fluid or blood exposure)
VACCINE SIDE EFFECTS	Local reaction (soreness, swelling and redness at injection site)
DOSING SCHEDULE	• Various forms: monovalent recombinant HB, HA/HB combination • Generally 3 doses administered at 0, 1 and 6 months • Age and schedule may vary by province (e.g. Ontario has 2-dose schedule in the 7th grade)

PREVENTION OF VIRAL HEPATITIS

Hepatitis A vaccines

- Inactivated vaccine
- Preferred for pre-exposure prophylaxis
- Used for postexposure prophylaxis
- Safe, well-tolerated
- Single or in combination with HBV vaccine

Hepatitis B vaccines

- Induces production of anti-HBs
- Response for adults decreases with age, highest response rate observed between 5 and 15 years of age
- Safe, well-tolerated
- Revaccination may be needed in hypo/non-responders (e.g. elderly, immunosuppressed)

Hepatitis B immune globulin (HBIG)

- Pooled Ig from donors to provide immediate short-term passive immunity
- Used in post-exposure prophylaxis
- Given with hepatitis vaccines at different injection site
- No effect on antibody response to vaccine

PREVENTION OF VIRAL HEPATITIS

Patient Characteristics (high risk individuals who should be vaccinated)	Hep A Vaccine	Hep B Vaccine
Infants and pre-adolescent children		X
People with end-stage renal disease or recipients of dialysis		X
People who receive repeated plasma-derived clotting factor replacement	X	X
People with HIV infections		X
People with chronic liver disease including HCV	X	X
Health care and emergency service workers who have exposure to blood in the workplace that may contain virus		X
People handling HAV-infected primates or involved in research on HAV	X	
People in training in school of dentistry, laboratory technology, medicine, nursing and other allied health professions		X
Illicit drug users of both injectable and non-injectable drugs	X	X

Legend:

x = Highly recommended

? = Recommended when there is evidence for sustained HAV transmission

PREVENTION OF VIRAL HEPATITIS

Patient Characteristics (high risk individuals who should be vaccinated)	Hep A Vaccine	Hep B Vaccine
People with multiple sexual partners, persons with a history of STIs		X
Men who have sex with men	X	X
Household and sexual contacts of people identified as HB-positive		X
Staff and inmates of long-term correctional facilities	v	X
Patients and staff of institutions for the developmentally challenged	v	X
International travellers to HA- or HB- endemic areas	X	X
Recipients of solid organ or stem cell transplantation		X
Individuals who emigrated from highly HBV-endemic areas		X

Legend:

x = Highly recommended

v = Recommended when there is evidence for sustained HAV transmission

HERPES ZOSTER (SHINGLES)

DISEASE SYMPTOMS AND COMPLICATIONS	Painful rash and blisters, often presents as a stripe of blisters around left or right side of torso, vision loss, postherpetic neuralgia (burning nerve pain even after shingles resolution)
DISEASE SPREAD	• Caused by reactivation of an initial varicella-zoster infection (chickenpox) • An individual who acquires an initial varicella infection will develop chickenpox, not shingles
VACCINE INDICATION	• Prevention of herpes zoster regardless of varicella infection history • 50 years of age or older • Recommended for immunocompromised individuals ≥ 18 years old
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• Shingrix® (NON-LIVE, recombinant zoster vaccine): 2 doses intramuscularly • Initial dose (month 0) followed by a second dose 2-6 months later ○ If needed, 2nd dose can be administered at least 4 weeks after 1st dose (e.g. for patients starting on immunosuppressive treatment soon)

After a shingles infection, when can one get the vaccine?

HAEMOPHILUS INFLUENZA TYPE B (HIB)

DISEASE SYMPTOMS AND COMPLICATIONS	Bacterial infection most commonly affecting children <5 years: meningitis, fever and chills, pneumonia, otitis media, developmental delays
DISEASE SPREAD	Respiratory droplets, cough, sneezing, close contact
VACCINE INDICATION	Routine immunization for prevention of Hib and associated complications
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• Usually with DTaP-IPV-Hib vaccine • Routine immunization of infant: at 2, 4, 6, and 12-23 months of age (4th dose usually given at 18 months) • Schedule may vary depending on province or territory

Note: After age 5, Hib is no longer required in healthy individuals. If an adult has not had DTaP-IPV-Hib, unless immunocompromised, they will receive DTaP and IPV only

INVASIVE MENINGOCOCCAL DISEASE

DISEASE SYMPTOMS AND COMPLICATIONS	Bacterial infection most commonly affecting children < 5 years; causes meningitis, sepsis, brain damage, neurologic disabilities, hearing impairment, amputations, death
DISEASE SPREAD	Person-to-person, mucosal contact with respiratory droplets from infected persons (e.g. coughing, sneezing)
VACCINE INDICATION	Routine immunization for prevention of invasive meningococcal disease
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• 3 different vaccines: Men-C, Men-B, and quadrivalent (A, C, Y, W-135) • Generally 2 or 3 doses depending on age of initial starting dose, (specific to provincial schedule, type of vaccine available)

HUMAN PAPILLOMAVIRUS

DISEASE SYMPTOMS AND COMPLICATIONS	• Typically self-limiting (i.e. resolves on its own) • May be associated with cervical and anogenital cancers • May cause non-cancerous lesions (e.g. genital warts, warts on oral mucosa and respiratory tract)
DISEASE SPREAD	• Sexually transmitted: direct skin or mucosa contact • Most infected individuals are asymptomatic, can spread virus to other unknowingly
VACCINE INDICATION	• Prevention of HPV and cervical cancer • Recommended for all persons 9 to 26 years, but older individuals may still get it
VACCINE SIDE EFFECTS	• Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• 3 different types of the vaccine: Bivalent (2vHPV), and <u>nine-valent (9vHPV: Gardasil 9®)</u> • Example for 9vHPV: → individuals 9-20 years of age: 1 dose schedule (a 2-dose schedule may be considered as an alternative) → individuals ≥ 21 years of age: 2 doses (at least 6 months apart) → Immunocompromised or HIV infection: 3 doses • Age of vaccination and schedule may vary depending on province or territory

JAPANESE ENCEPHALITIS

DISEASE SYMPTOMS AND COMPLICATIONS	• Often asymptomatic, flu-like symptoms, confusion, coma and neuropsychiatric complications, death
DISEASE SPREAD	• Bite of infected mosquito (RNA virus from <i>Flaviviridae</i> family) • Rural areas of Southeast Asia
VACCINE INDICATION	• Prevention of infection with virus • Individuals travelling to endemic areas and are at higher risk of disease
VACCINE SIDE EFFECTS	• Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• 2 doses (days 0 and 28); accelerated schedule (days 0 and 7) possible for those between 18-65 years of age • Last dose should be at least 1 week prior to potential exposure • Booster should be given if travelling again or continue to be at risk of exposure 1 year after previous dose

PNEUMOCOCCAL INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	Pneumonia, sinus and/or ear infection, meningitis, septicemia, deafness, death
DISEASE SPREAD	Respiratory droplets, sneezing, coughing, saliva; direct oral contact or or indirect contact with the respiratory secretions of infected individual
VACCINE INDICATION	Prevention of <i>Streptococcus pneumoniae</i> infection and complications
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• Pneumococcal Conjugate 13-valent Vaccine (Pneu-C-13; Prevnar®13) • Pneumococcal Conjugate 15-valent Vaccine (Pneu-C-15; Vaxneuvance®) • Pneumococcal Conjugate 20-valent Vaccine (Pneu-C-20; Prevnar®20) • Pneumococcal Polysaccharide 23-valent Vaccine (Pneu-P-23; Pneumovax®23) • Pneumococcal Conjugate 21-valent Vaccine (Pneu-C-21) *NEW*

CHILDHOOD IMMUNIZATION – PNEUMOCOCCAL VACCINE

FYI

OLD recommendations (No need to know)

Healthy infants (2 to less than 12 months of age)

- Pneu-C-13 vaccine 3 dose schedule: 2 months, 4 months and 12 months of age

OR

- Pneu-C-13 vaccine 4 dose schedule: 2 months, 4 months and 6 months followed by a dose at 12 to 15 months of age

CHILDHOOD IMMUNIZATION – **NEW** PNEUMOCOCCAL VACCINES

Healthy infants (2 to less than 12 months of age)

2 months to <7 months



Choice of vaccine: Pneu-C-15 or Pneu-C-20
→ 3 doses at: 2, 4, and 12 months of age
OR
→ 4 doses at: 2, 4, 6, and 12-15 months of age

7 months to < 12 months



Choice of vaccine: Pneu-C-15 or Pneu-C-20
→ 2 doses of either vaccine, 8 weeks apart,
followed by a 3rd dose 12-15 months of age

Healthy infants (12 months+)



→ Assessment required to determine # of doses needed to complete the immunization series
→ Pneumococcal vaccine not recommended for children past 5 years of age

Children who have completed Pneu-C-13 series do not need an additional dose of the newer vaccines

ADULT IMMUNIZATION – PNEUMOCOCCAL VACCINE

No Prior Vaccination

A single dose of Pneu-C-20 OR Pneu-C-21 can be given to:

- Adults aged 65+
- All adults aged 50-64 with risk factors (e.g. chronic heart disease, chronic liver disease, chronic kidney disease, chronic lung disease, diabetes mellitus, immunocompromised)
- Adults aged 18-49 who are immunocompromised (e.g. sickle cell anemia, HIV, immunocompromising therapy)

Prior Vaccination and aged 65+

Previous immunization with Pneu-P-23, Pneu-C-13 or Pneu-C-15

Give a single dose of Pneu-C-20 or Pneu-C-21 at least 1 year* after last dose

A series of Pneu-C-15 followed by Pneu-P-23 (1 year later) may be considered as alternative if Pneu-C-20 or Pneu-C-21 are not available

* While a 1-year interval is recommended and preferred, a minimum interval of 8 weeks may be considered in those starting on immunosuppressive treatment or who have diseases that may cause immunodeficiency

RABIES INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	• Flu-like symptoms • Fatal CNS infection • Paralysis
DISEASE SPREAD	• Saliva of infected animal • Disease of mammals: dogs, skunks, raccoons, bats, foxes • Enters through open skin: bites, broken skin, scratches, respiratory tract, or mucous membranes
VACCINE INDICATION	• Prevention and post-exposure against rabies • Those working with animals or travelling to endemic areas
VACCINE SIDE EFFECTS	• Soreness, swelling, redness at injection site, headache
DOSING SCHEDULE	• Pre-exposure (prevention): 3 doses (day 0, day 7, and day 21-28) for immunocompetent and high/very high risk, or immunocompromised. 2 doses for immunocompetent and moderate risk when risk is expected to be < 3 years • Post-exposure prophylaxis (never vaccinated): 4-5 doses, depending on whether immunocompetent or immunosuppressed • Post-exposure prophylaxis (previously vaccinated): 2 doses

MEASLES, MUMPS, AND RUBELLA (MMR)

***Not administered by pharmacists**

DISEASE SYMPTOMS AND COMPLICATIONS (Measles)	• Fever, cough, Koplik spots, conjunctivitis, rash, runny nose
DISEASE SPREAD (Measles)	• Airborne virus, respiratory droplets, direct contact with throat/nasal secretions
VACCINE INDICATION	• Prevention of measles, mumps, and rubella
VACCINE SIDE EFFECTS	• Soreness, swelling and redness at injection site, rash, headache
DOSING SCHEDULE	• MMRV and MMR • Childhood immunization: 2 doses (first dose at 12-15 months, second dose at 18 months) • Adults born during or after 1970: 1 dose or 2 doses in those with high risk of exposure (e.g. health care workers, military personnel)

VARICELLA INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	• Fever, itching, small blisters, scabs, bacterial infections, pneumonia, encephalitis
DISEASE SPREAD	• Airborne virus, sneezing, coughing, fluid from blisters, saliva
VACCINE INDICATION	• Prevention of varicella zoster virus infection • Routine immunization (ages 12 months to 13 years old)
VACCINE SIDE EFFECTS	• Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• MMRV and univalent • 2 doses: schedule and age of immunization depend on provincial or territorial routine vaccination schedules • Recommended to AVOID use of salicylates for 6 weeks after vaccination

TYPHOID DISEASE (TYPHOID FEVER)

DISEASE SYMPTOMS AND COMPLICATIONS	Flu-like symptoms, GI symptoms, abdominal perforation, extra-intestinal infection, death
DISEASE SPREAD	Fecal-oral route: ingestion of food or water that is contaminated with feces of infected individuals
VACCINE INDICATION	Prevention of infection by <i>Salmonella typhi</i> , travelling to endemic region
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• Oral (live): four doses - one capsule every other day for a week (day 1, day 3, day 5, and day 7); the last dose should be given at least 1 week before travel; swallow capsule (NOT chew) 1 hour prior or 2 hours after a meal with a cold or lukewarm drink; booster is needed every 7 years for those who remain at risk • IM (inactive): one dose given at least 2 weeks before travel; booster needed every 3 years for those remaining at risk

YELLOW FEVER INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	Flu-like symptoms, jaundice, photophobia, epigastric pain, renal failure, thrombocytopenia and hemorrhagic symptoms
DISEASE SPREAD	Mosquito bite: Primarily mosquitoes of the <i>Aedes</i> or <i>Haemogogus</i> species
VACCINE INDICATION	Prevention of infection, traveling to endemic region
VACCINE SIDE EFFECTS	Soreness, swelling and redness at injection site, headache
DOSING SCHEDULE	• 1 dose (ideally 10 days before travel) • Re-immunization is recommended for certain individuals after 10 years (controversy over the need for this)
Contraindications	• History of thymus disease with abnormal immune function (e.g. myasthenia gravis, thymoma or prior thymectomy) • Immunodeficiencies (e.g. immunosuppressive therapies or conditions)

TUBERCULOSIS

DISEASE SYMPTOMS AND COMPLICATIONS	Infection is usually latent, symptoms of active disease include cough, weight loss, fever, night sweats
DISEASE SPREAD	• Airborne transmission (e.g. inhalation of infected droplets) • Extended exposure (weeks) with infected person usually required for transmission
VACCINE INDICATION	• Prevention of <i>Mycobacterium tuberculosis</i> infection • Infants born in high-risk areas • NOT recommended for routine use in Canadians but may be considered in exceptional circumstances (e.g. long-term travellers to endemic areas) • Vaccine: Bacille Calmette-Guérin (BCG)
DOSING SCHEDULE	Single dose

INFLUENZA INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	• Fever, fatigue, cough, headache, muscle pain • Pneumonia (bacterial or viral) and other infections (e.g. sinus) • Hospitalization, death
DISEASE SPREAD	Sneezing, coughing, direct/indirect contact with respiratory secretions
VACCINE INDICATION	All individuals $\geq$ 6 months of age
VACCINE SIDE EFFECTS	Soreness, swelling, and redness at injection site, headache
DOSING SCHEDULE	• 1 single dose. Re-immunization is recommended annually • Children aged 6 months to < 9 years receiving vaccine for the <u>first</u> time: 2 doses, 4 weeks apart • Live attenuated influenza vaccine (LAIV) is contraindicated in pregnancy, immunocompromised patients, children taking acetylsalicylic acid etc. • Can be used in patients with egg allergies • Types: recombinant, adjuvanted trivalent, quadrivalent, high-dose quadrivalent, and live attenuated influenza vaccine (given intranasally) • Vaccine recommended to be taken before November (2 weeks to develop immune response)

TYPES OF INFLUENZA VACCINE

- Inactivated influenza vaccines (IIV)
 - Standard-dose, egg-based, quadrivalent inactivated influenza vaccine (IIV4-SD)
 - Standard-dose mammalian cell culture-based quadrivalent inactivated influenza vaccine (IIV4-cc)
 - Adjuvanted inactivated trivalent influenza vaccine (IIV3-Adj)
 - High-dose inactivated quadrivalent influenza vaccine (IIV4-HD)
- Recombinant influenza vaccine (RIV)
- Live attenuated influenza vaccine (LAIV)

INFLUENZA VACCINE

Children between ages of 6 months to 9 years receiving vaccine for first time require two doses spaced out for a minimum of at least 4 weeks. All other age groups require only one dose

Age group	IIV4-SD	LAIV4	IIV4-HD	IIV3-Adj	IIV4-cc	RIV4
6 – 23 months	Yes			Yes	Yes	
2 – 17 years	Yes	Yes			Yes	
18 – 59 years	Yes	Yes			Yes	Yes
60 – 64 years	Yes				Yes	Yes
65+ years	Yes		Yes	Yes	Yes	Yes
Pregnant women	Yes				Yes	Yes

WHAT ARE THE PROPERTIES OF TRAVELLER’S DIARRHEA AND HOW CAN IT BE PREVENTED?

Disease symptoms and complications	Loose stools, nausea, vomiting, fever, abdominal cramps lasting up to 5 days (typically self-resolves)		
Disease spread	Foods in endemic areas: fecal-oral contact by eating raw fruits, vegetables, untreated water		
Vaccine indication	Prevention of travelers' diarrhea and cholera caused by <i>Enterotoxigenic E. coli</i> /ETEC (efficacy: 67%) and <i>V. cholerae</i> (efficacy: 85%)		
Dosing schedule	- Dukoral® PO vaccine that is stored in the fridge - Do not eat, drink, or take any medicine 1 hour before or after vaccine		
		Cholera prevention	ETEC diarrhea prevention
	Ages 2 to <6 years	- Take 3 oral doses at least 7 days apart; last dose should be taken 1 week before travel date - Booster: one booster dose taken 1 week before travel date if last dose was between 6 months - 5 years ago	Take 2 oral doses at least 7 days apart. Last dose should be taken 1 weeks before travel
	Age ≥6 years	- Take 2 oral doses at least 7 days apart. Last dose should be taken 1 week before travel date - Booster: 1 booster dose taken 1 week before travel date if last dose was between 2-5 years ago	Booster: 1 booster dose taken 1 week before travel date if last dose was between 3 months to 5 years ago

RESPIRATORY SYNCYTIAL VIRUS (RSV) INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	• Cold-like symptoms, nasal congestion, cough, low grade fever, loss of appetite. Usually resolves in 1-2 weeks • May develop into bronchiolitis and pneumonia. Can result in hospitalizations and ICU admission if severe
DISEASE SPREAD	• Direct and indirect contact with infectious secretions e.g. respiratory droplets through cough and sneezing, touching contaminated surfaces
VACCINE INDICATION (Arexvy®, Abrysvo® and mRESVIA®)	Prevention of LRT disease caused by RSV • Recommended for ≥ 75 years old, and ≥ 60 years in nursing homes and LTC facilities. Younger individuals (50-74 years old) may receive the vaccine in consultation with their HCP • Abrysvo® is the only one indicated in pregnant women (from 32-36 weeks of gestation) → not necessary if infant is expected to receive nirsevimab due to superior efficacy • Nirsevimab: monoclonal antibody recommended for infants < 8 months entering or born during their first RSV season, or for infants at high risk of severe disease
VACCINE SIDE EFFECTS	• Injection-site reactions, headache, fatigue
DOSING SCHEDULE	• Vaccines: Single dose IM injection (0.5 mL) administered just before the start of the RSV season

TETANUS, DIPHTHERIA and PERTUSSIS INFECTION

DISEASE SYMPTOMS AND COMPLICATIONS	• Pertussis ('whooping cough'): initial symptoms similar to flu/cold, progresses to coughing fits → can lead to SOB, choking, vomiting, brain damage • Tetanus ('lockjaw'): muscle spasms beginning in neck and face, trouble breathing and swallowing → can lead to respiratory failure, seizures • Diphtheria: severe throat soreness (think grey membrane), swollen neck glands → can lead to airway obstructions, CV complications, nerve damage
DISEASE SPREAD	• Pertussis and Diphtheria: contagious; inhalation of respiratory droplets • Tetanus: not contagious; spores enter through broken skin contaminated with soil, feces or dust
VACCINE INDICATION	Prevention of tetanus, diphtheria and pertussis infections
VACCINE SIDE EFFECTS	Soreness, swelling, and redness at injection site, headache
DOSING SCHEDULE	• Routine immunization in infants and children • <i>Every</i> pregnancy, regardless of prior immunization (preferably between 27-32 weeks of gestation) • Adults should receive the vaccine if not previously vaccinated in adulthood, and a booster of tetanus and diphtheria every 10 years • Tetanus vaccine alone may be needed for post-exposure/wound management - depends on the nature of the wound and vaccination history

CASE 1

GE, a 66-year-old patient comes into the pharmacy with a prescription for Shingrix® (recombinant zoster vaccine for shingles) and wants more information about it. GE has a past medical history significant for type II diabetes (T2DM), hypertension, and a history of chickenpox. He currently takes ramipril 2.5 mg PO daily, rosuvastatin 20 mg PO HS, and metformin 500 mg PO BID. He has no allergies to medications. He weighs 140 lbs (BMI:22 kg/m²) and his most recent HbA1C was 7%.



CASE 1 QUESTIONS

1. Which of the following is false?

- a) Shingrix® is administered in 2 doses at 2-6 months apart
- b) He's at risk for shingles due to his age, T2DM and history of chickenpox
- c) Shingles is a contagious disease
- d) You can offer to administer it as a pharmacist

2. How should Shingrix® be administered?

- a) SC at 45 degrees with 5/8" 23G needle
- b) SC at 90 degrees with 5/8" 25G needle
- c) IM at 90 degrees with 1" 25G needle
- d) IM at 90 degrees with 1½ " 27G needle

3. Which of the following vaccines would you also strongly recommend?

- a) Live attenuated influenza
- b) Pneumococcal
- c) MMR
- d) Hepatitis A



CASE 2

EM is a 30-year-old patient who is currently 8 months pregnant. She comes into your pharmacy asking whether she should get the flu shot now that she is pregnant. EM is in good health but has a history of an egg allergy. EM is currently taking Pregvit® 1 tab PO BID.



CASE 2 QUESTIONS

1. What should you advise EM regarding the influenza vaccine?
 - a) It's not recommended during pregnancy
 - b) It's not recommended due to her egg allergy
 - c) Recommend to administer LAIV4
 - d) Recommend to administer IIV4-SD

2. Ten-minutes into observation, EM reports of some pain and swelling at the injection site. What should you do?
 - a) Immediately administer epinephrine and call 911
 - b) Do nothing and observe for up to 30 minutes
 - c) Give her diphenhydramine and send her to the emergency room
 - d) Tell her not to get the influenza vaccine again

3. One year later, EM comes with her child to the pharmacy for his first flu shot. Which of the following is appropriate?
 - a) Administer the live attenuated vaccine
 - b) Administer 2 live attenuated vaccine doses 4 weeks apart
 - c) Send EM's son to his family doctor to receive the flu shot
 - d) Administer 2 standard dose quadrivalent vaccine doses 4 weeks apart



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CHANGE LOG

August 2025

- Formatting changes made throughout
- Slide 40: Added Pneu-C-21 option to the tables as per Health Canada recommendations. Also updated interval between Pneu-P-23 and Pneu-C-20 or C-21 to 1 year instead of 5 years

October 2025

- Slide 18: Added information to footnote for IM administration to the deltoid muscle in adults and adolescents (>12 years)

November 2025

- Formatting changes made throughout
- Slide 17: Added child consent statement and examples for life-saving injections
- Slide 27: Inserted ideal first dose timing
- Slide 40: Simplified flow chart for adults 65+ with prior vaccination, added footnote on minimum interval between Pneu-C-vaccines
- Slide 45: Added window of vaccination
- Slide 47: Added information on children < 9 years receiving flu shot for the first time

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 3: Updated legend
- Slide 32: Updated vaccine indication and dosing schedule for Herpes Zoster
- Slide 41: Updated dosing for pre-exposure as per updated 2026 guidance
- Slide 51: Added new slide on RSV immunization
- Slide 52: Added tetanus, diphtheria and pertussis immunization
- Slide 58: Updated references